Activity 4 - Define Organisational and Product Lifecycle Boundaries

Alignment: K3 | K5 | A1 | A2 | LO2 Duration: 50 minutes Objective: Apply consolidation, reporting-boundary and lifecycle rules to HarbourLight's organisation and Lumina 400 product.

Scenario

HarbourLight owns the Tuas assembly plant, has a 45% stake in a Batam driver-board venture without operating authority, leases a Singapore office, uses a third-party Johor warehouse and sells panels across ASEAN. The Lumina 400 has aluminium housing, imported chips and drivers, a 50,000-hour rated life, variable regional electricity use and end-of-life recycling claims with limited evidence.

Materials

- Entity and control fact sheet
- Lumina 400 lifecycle and configuration data
- Customer-use and end-of-life assumptions
- Boundary decision workbook

Questions

- Which operations are included under the selected consolidation approach?
- What function and product configuration are assessed?
- Which lifecycle stages and flows are included or excluded?
- Which assumptions could materially change the result?

Detailed procedure

1. State the organisational-inventory purpose, product-footprint purpose, intended users and reporting period.
2. Map all entities, facilities, leases and outsourced operations with ownership and control evidence.
3. Compare equity share, financial control and operational control; select and justify one consolidation approach.
4. Apply the approach consistently and record included, excluded and unresolved organisational items.
5. Define the Lumina 400 functional unit including output, rated life and performance assumptions.
6. Map cradle-to-grave stages from raw materials through component manufacture, assembly, distribution, use and end of life.
7. Set cut-offs, allocation, recycled-content, use-profile and end-of-life rules with evidence or transparent assumptions.
8. Test boundary decisions against relevance, completeness, consistency and decision-usefulness.
9. Peer-challenge one exclusion and record the final disposition and owner.

Required evidence

- Organisational boundary diagram
- Lifecycle process map

- Functional unit and system boundary
- Assumption and exclusion log

Acceptance check

Every named operation and lifecycle stage is addressed; the functional unit is comparable and measurable; exclusions and assumptions have transparent rationale, evidence and ownership.

Files

- Activity-04-Workbook.xlsx - editable working file
- Activity-04-Activity-Pack.pdf - printable scenario, questions, procedure and checklist

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